ARM-450 — Slicing Specification

Read the two constraints in §1 before anything else. They are not preferences; the parts do not assemble if you get them wrong.

Generated from cad/params.py, which now asserts both of them at import.

Material: plain PLA (params.MATERIAL = "PLA"). E = 3500 MPa, 1.24 g/cm³, yield 55 MPa. Every load case was re-run against it — 10 of 10 pass, with the worst-case deflection 0.197 mm against a 0.30 mm limit. Backlash still dominates at 2.25 mm, so the material change is invisible at the tool.

> If a print is running now, check the perimeter count before anything else. This project's parameter file said NOZZLE = 0.6 with 4 perimeters until 2026-09-02, while the machine in use has a 0.4 mm nozzle. Four perimeters at a 0.48 mm line width is 1.92 mm against a 2.4 mm structural wall — every wall in the arm prints with a 0.48 mm void up its middle, filled with sparse infill, and looks perfect from the outside. Five perimeters at 0.48 mm consume the wall exactly. The nozzle is now the single input and the line width and perimeter count are derived from it.

1. The two settings that are not yours to choose

Layer height = 0.20 mm

The link seam is a tongue and groove. The tongue is 1.60 mm and must be an integer number of layers, because a slicer rounds a partial layer up:

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1.60 / 0.20 = 8         exactly            USE THIS
1.60 / 0.16 = 10        exactly            also valid
1.60 / 0.32 = 5         exactly            also valid, coarse
1.60 / 0.25 = 6.4       rounds to 7 -> 1.75 mm tongue
1.60 / 0.28 = 5.71      rounds to 6 -> 1.68 mm tongue
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An over-tall tongue bottoms out in the groove before the two rims can touch, so the seam sits open and the closed box section — the whole reason the links are a clamshell — never closes. This already happened once at a 1.50 mm tongue, which is why it is 1.60 now.

Nozzle = 0.6 mm, hardened steel, 4 perimeters

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4 perimeters x 0.62 mm line = 2.48 mm  >=  the 2.40 mm design wall
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The wall is entirely consumed by perimeters, which is what makes it print solid, and that is the assumption behind the whole mass model (MASS_FILL = 0.90). Drop to 3 perimeters and you get 1.86 mm of wall with a void behind it — lighter than modelled, and weaker in exactly the direction the links are loaded.

2026-08-27 — the build material is now PLAIN PLA, not carbon-filled, and that removes the nozzle problem entirely. Carbon fill is abrasive and eats a brass nozzle inside a spool; plain PLA does not. A standard brass nozzle is fine, in 0.4 or 0.6. No hardened nozzle to buy, no clogging, and layer adhesion is actually better — carbon fibres interrupt the bond between layers.

If you only have a 0.4 mm nozzle

It works — but not at the default line width, and that is the whole catch. At 0.45 mm you get five perimeters totalling 2.25 mm and a 0.15 mm gap the slicer fills with infill or simply leaves open, which breaks the solid-wall assumption the mass model and the stiffness numbers both rest on.

Two settings consume the 2.40 mm wall exactly:

nozzle	line width	perimeters	total	
0.6	0.60	4	2.40	the shipped profile
0.4	0.48	5	2.40	use this — fewer passes, so faster
0.4	0.40	6	2.40	alternative: narrower lines follow the Ø42 bore better

0.48 is 1.2× the nozzle, which is the normal upper limit and perfectly standard.

Everything else stays the same. Layer height is still 0.20 mm — it divides the 1.6 mm tongue into 8, and that is independent of nozzle. (0.16 and 0.32 also divide it; 0.32 is the practical ceiling for a 0.4 nozzle if you want the time back.)

One thing gets better. The minimum printable wall is two lines, so it falls from 1.24 mm to 0.90 mm — the 2 mm gear teeth, the bayonet lugs and the raised coupon marks all get easier, not harder.

One thing gets worse: time. Wall passes scale with 1/line-width, so the full set goes from roughly 21 h to 29 h.

> ### Two warnings, and the first is not optional

> The nozzle must be HARDENED STEEL, in any diameter. Carbon fill opens a brass nozzle measurably inside one spool. A nozzle that creeps from 0.40 to 0.45 mid-build changes every wall and every bearing pocket on the parts printed after it — and nothing warns you. You would find out at assembly, with half the parts fitting and half not.

> Carbon fill clogs a 0.4 mm bore far more readily than a 0.6. Many CF filaments specify 0.5 mm minimum for exactly this reason. It is survivable — dry filament, slower retraction, higher temperature — but it is the failure mode you will actually meet.

> A 0.6 mm hardened nozzle is the cheapest line item in this entire build, and it removes the clog risk, the wear risk and eight hours of print time at once. If you can buy one before starting, do.

2. Profile

setting	value	why
Layer height	0.20 mm	§1 — integer tongue
First layer	0.25 mm	adhesion
Nozzle	0.6 mm or 0.4 mm, brass is fine	plain PLA is not abrasive
Wall line width	0.48 mm (0.4 nozzle — the machine in use) · 0.60 mm on a 0.6	must divide the wall exactly
Perimeters	5 (0.4 nozzle) · 4 on a 0.6	§1 — 5 × 0.48 = 2.40 consumes the wall EXACTLY
Top / bottom layers	5 / 4	1.00 / 0.80 mm solid
Infill	40 % gyroid	see §3 for the exceptions
Infill overlap	15 %	bonds infill to the solid wall
Nozzle temp	200–215 °C	plain PLA; CF would want 225–235
Bed	60 °C	
Cooling	100 % after layer 3	
Outer wall speed	25 mm/s	dimensional accuracy on the Ø42 pockets
Inner wall / infill	60 / 80 mm/s	
Wall order	inner → outer	the outer wall lands on solid support
Z-seam	Aligned, rear	never Random — see §4
XY compensation	0.00 mm to start	the coupon sets this — expect to need it, §5
Ironing	OFF	it smears the pocket mouth
Supports	none	every part is oriented to avoid them, §3
Brim	5 mm on tall/narrow parts	§3

Dry the filament if it has been open a while. Plain PLA is far less hygroscopic than CF-filled, so this matters less than it did — but a wet-printed furry wall is still an undersized bearing pocket.

3. Per part — orientation, infill, brim

Orientation is not optional. Each one puts the loaded direction across layers rather than along them, and keeps overhangs off surfaces that matter.

part	qty	orientation	infill	brim
link_half_tongue	2	split face DOWN, open side up	40 %	no
link_half_groove	2	split face DOWN, open side up	40 %	no
wrist_j4_housing	1	bearing bore axis VERTICAL	50 %	no
wrist_j5_yoke	1	bearing bore axis VERTICAL	50 %	yes
wrist_j6_output	1	bearing bore axis VERTICAL	50 %	no
turret_j1	1	bearing seat DOWN, axis vertical	40 %	yes
base	1	foot DOWN, upright	30 %	no
shaft_clamp	2	bore axis VERTICAL	60 %	no
servo_collar	2	collar axis VERTICAL	60 %	no
horn_adapter	6	horn face DOWN on the bed	100 %	no
fit_coupon	1	flat, pockets facing UP	40 %	no
tool_adapter	1	spigot UP, flat face on the bed	60 %	no
tool_gripper	1	socket face UP	40 %	no
gripper_jaw	2	rack bar FLAT on the bed, teeth sideways	100 %	yes
gripper_pinion	1	flat, teeth in the XY plane	100 %	no
tool_dock	1	cone DOWN on the bed, socket up	40 %	no
dock_target	1	flat face on the bed, spigot up	40 %	no
dock_target_sealed	1	flat face on the bed, spigot up	40 %	no

100 % infill where it is called for is not caution. The pinion teeth are 2.0 mm and the rack teeth 2.4 mm deep — at 40 % they are hollow shells that shear off the first time the jaws grip. The horn adapter is 100 % because it is the part that carries servo stall torque into the shaft; it is small, so it costs nothing.

dock_target and dock_target_sealed are alternatives, not a pair. The plain one is the dry docking demonstration; the sealed one carries a Ø14.40 × 1.30 O-ring groove in the spigot for fluid transfer. Print whichever the demonstration needs — they are otherwise identical and both dock the same way.

> The O-ring groove is 0.45 mm deep and 1.30 mm wide, cut into a Ø15.30 spigot. Print it spigot up so the groove is a horizontal feature — printed on its side it becomes a stack of stair-steps that the O-ring cannot seal against. Slow the outer wall for that part; it is the one printed surface in the design that has to be fluid-tight.

Brim where listed: gripper_jaw is 46 × 8 mm on its footprint — a tall thin fin. turret_j1 is 80 × 80 × 81 and wrist_j5_yoke is 42 mm tall on a small base.

No supports anywhere. The docking probe's capture cone prints cone down, which makes it a 45° self-supporting overhang. Printing it the other way up needs support inside the cone, and removing that support leaves exactly the surface finish the cone exists to avoid.

4. Why the Z-seam setting matters here

Set it to Aligned and park it at the rear. On Random, the slicer drops a seam wherever it likes, and on a part like wrist_j5_yoke that means somewhere on the Ø37 bearing pocket wall. A seam blob inside a bearing seat is a high spot on the one surface whose diameter you are trying to control to ±0.05 mm.

5. Print the fit coupon first — 0.9 h, and it gates 19 h

The coupon carries three pockets at Ø41.95 / Ø42.00 / Ø42.05, three insert-boss hole sizes and a seam sample. Every feature is labelled on the part: raised numerals on the top strip, and 1 / 2 / 3 raised dots beside each feature that read the same way up or down.

Print it with the profile above, unchanged. That is the point — it measures your machine running the settings you will use for the real parts, so changing anything afterwards invalidates it.

> ### Plain PLA shrinks nearly 3× more, and it matters here

> | | shrink | on a Ø42 bore | | --- | --- | --- | | PLA+CF | 0.15 % | 0.06 mm | | plain PLA | 0.40 % | 0.17 mm |

> The coupon brackets only 0.10 mm (Ø41.95 → Ø42.05), and plain PLA's shrink is larger than that whole bracket. There is a real chance all three pockets come out undersize and none of them presses correctly.

> That is the coupon working, not failing. If all three are too tight, set XY compensation to about −0.15 mm, reprint the coupon, and read it again.

Budget two coupon prints, not one. 0.9 h each, and it still gates 19 h of parts.

Reading it:

result	means
firm thumb press, seats square	correct — use that diameter
drops in freely	pocket is oversize → set XY compensation negative by half the step
will not start	pocket is undersize → XY compensation positive

Then set BRG_FIT in cad/params.py to the winning step, re-export, and print the structural parts. If Ø42.00 wins, nothing changes and you print as-is.

Do not skip to the real parts. Roughly 19 of the 21 print hours are parts carrying a Ø42 or Ø37 pocket. If the fit is wrong they are all scrap, and the coupon costs 0.9 h and 43 g to find out.

6. Filament budget

788 g of PLA+CF for the complete set at the infills above — one 1 kg spool, with about 210 g spare. That is not much margin for a failed print, so if you can, buy two.

7. Assembly notes that depend on the print

- Install the heat-set inserts while the parts are fresh, 200–220 °C, pressed slowly

and square. The coupon's three boss holes tell you which diameter takes them without splitting the boss.

- The seam screws are M3 × 20 (usable 18.0–22.0) and the tip ears M2.5 × 18

(usable 16.9–19.5). Too short and they never reach the insert; too long and they bottom in the blind hole and jack the halves apart.

- Heads sit proud on the outer face — there is no counterbore, deliberately.